

Magma Series & Geodynamic Environments: The Calc-Alkaline Subduction Hallmark & Petrogenetic Evolution

Comprehensive Igneous Petrology Guide · Authoritative Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Ch. 8, 16 & 17)

THEORETICAL FOUNDATION: MAGMA SERIES & DISCRIMINATION DIAGRAMS

Alkaline vs. Subalkaline, Tholeiitic vs. Calc-Alkaline, AFM Trends & Subduction Zone Petrogenesis

1. The Concept of Magma Series & The Two Major Splits

A **magma series** (or association, lineage, clan) is defined as a group of igneous rocks that share distinctive chemical and mineralogical characteristics and exhibit a coherent, continuous evolutionary trend on variation diagrams, reflecting descent from a common parental magma type via systematic differentiation:

The Two Classical Branching Points:

- First Split (TAS Diagram — Total Alkalis vs. Silica):** Magmas split into *Alkaline* (silica-undersaturated, high $\text{Na}_2\text{O} + \text{K}_2\text{O}$) and *Subalkaline* (silica-saturated to oversaturated). They are separated by the "thermal divide" across the basalt tetrahedron.
- Second Split (Subalkaline Division — AFM & Miyashiro Diagrams):** Subalkaline magmas further bifurcate into two fundamental evolutionary paths:
 - **Tholeiitic Series:** Characterized by strong initial *iron enrichment* (FeO^*) at nearly constant SiO_2 .
 - **Calc-Alkaline Series:** Characterized by continuous *silica enrichment* without iron enrichment.

"First, calc-alkaline magmas are essentially restricted to subduction-related plate tectonic processes. This must carry some genetic significance and will certainly be a factor in any model for subduction zone magma genesis. Recognition of calc-alkaline characteristics in the composition of ancient volcanic rocks may help determine their tectonic framework."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., 2010), Chapter 8: Major and Minor Elements in Magmas, p. 161.

2. Discrimination Diagrams: AFM and Miyashiro (FeO^*/MgO)

The standard triangular **AFM diagram** ($A = \text{Na}_2\text{O} + \text{K}_2\text{O}$, $F = \text{FeO} + 0.8998\text{Fe}_2\text{O}_3$, $M = \text{MgO}$) provides the classic graphical separation between tholeiitic and calc-alkaline magmas:

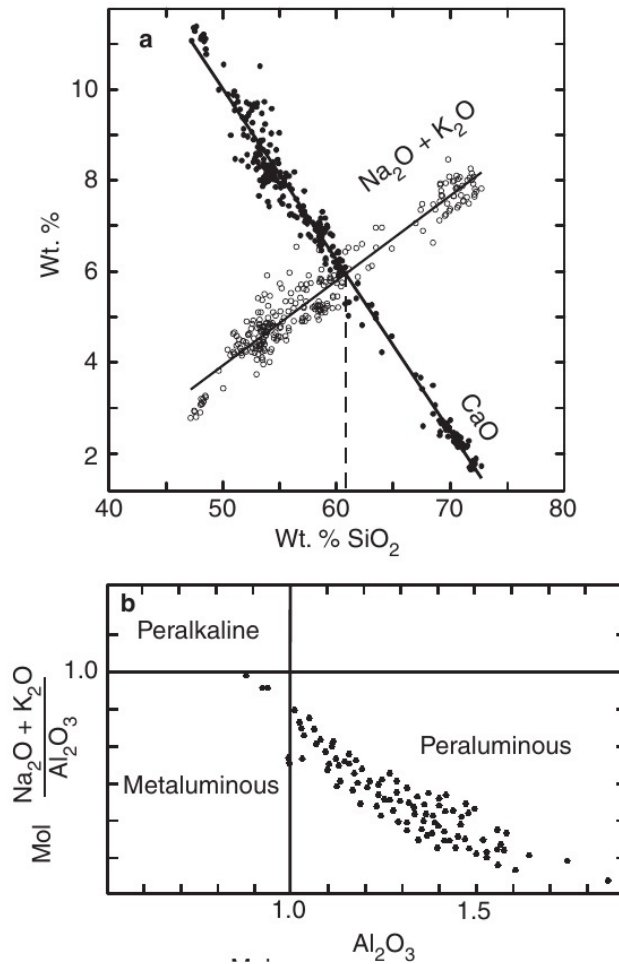


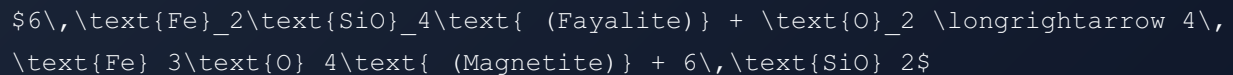
Figure 1 (Winter Fig. 14, p. 160): AFM diagram showing the contrasting differentiation trends. The solid line divides the upper **Tholeiitic field** from the lower **Calc-Alkaline field** (Irvine and Baragar, 1971). Tholeiitic suites (such as the Skaergaard intrusion, Mid-Atlantic Ridge MORB, and Thingmuli volcano, Iceland—solid circles) arch strongly toward the F apex (iron enrichment). Calc-alkaline suites (such as the Cascade volcanic arc and Crater Lake, Oregon—open circles) bypass the F apex, trending directly from the M corner toward the alkali A apex.

THE PHYSICAL CAUSE OF THE CALC-ALKALINE TREND: OXYGEN FUGACITY (f_{O_2}) & WATER

Why does the Calc-Alkaline series avoid iron enrichment?

- **Tholeiitic Differentiation (Anhydrous / Low f_{O_2}):** Under dry, reducing mantle conditions (e.g. MORB), fractional crystallization of olivine (Mg_2SiO_4) and plagioclase removes Mg^{2+} , Ca^{2+} , and Al^{3+} . Iron (Fe^{2+}) does not enter plagioclase and is rejected into the melt. Magnetite cannot crystallize early because oxygen is low. Thus, the residual liquid undergoes dramatic iron accumulation (FeO^*/MgO spikes).

- **Calc-Alkaline Differentiation (Hydrous / High f_{O_2}):** In subduction zones, water and oxidizing species (SO_4^{2-} , Fe^{3+}) released from the down-going slab elevate $P_{\text{H}_2\text{O}}$ and oxygen fugacity in the mantle wedge. High f_{O_2} forces iron into the ferric state (Fe^{3+}), triggering **early precipitation of titanomagnetite** (Fe_3O_4):



Early magnetite removal strips iron out of the melt, driving liquid composition directly toward silica enrichment ($\text{SiO}_2 > 56\%$, andesite to dacite to rhyolite).

ied in
espite
lds in
 Al_2O_3
lagio-

3 and
nd the
along
istinc-
ie two
s. The
aking
matic.
ble to
ie two
hment
e 14).
nd 15

both CaO and alkalis (as the original Peacock criteria do).

Irvine and Baragar (1971) further divided the alkali series into a **sodic series** and **potassic series**, and Middlemost (1975) recommended adding a **high-K series** as well (Figure 16). The remaining terms, such as *peraluminous*, *metaluminous*, etc., although they may yet be shown to characterize some distinctive magma series, are more useful as descriptive terms when some chemical characteristic is emphasized for a particular igneous rock or province.

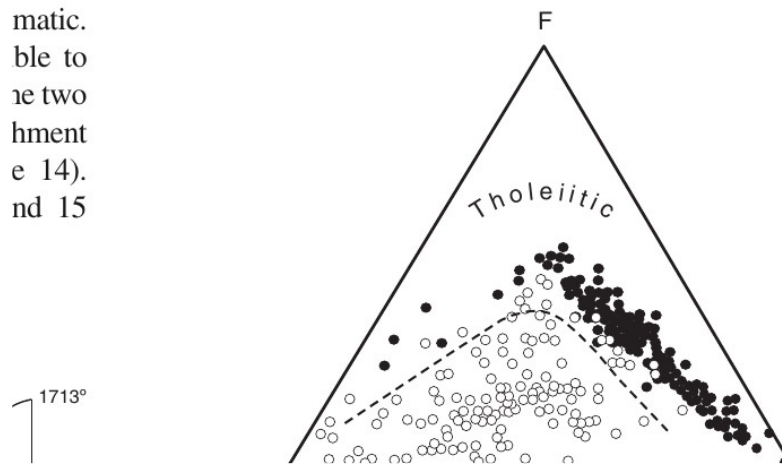


Figure 2 (Miyashiro Diagram): Plot of FeO^*/MgO vs. SiO_2 (Miyashiro, 1974). Tholeiitic magmas show a steep positive slope (FeO^*/MgO increases rapidly with little silica increase). Calc-alkaline magmas show a gentle, flat slope because iron is simultaneously removed by early Fe-Ti oxide fractionation.

3. Petrogenesis in Subduction Zones: Why Subduction is Essential

Winter's subduction zone model demonstrates why calc-alkaline magmatism cannot occur at mid-ocean ridges or intraplate hotspots:

cons of magmas at mid-ocean ridges.

Magma series are important to our understanding of igneous petrogenesis. Earlier, as we first regarded the spectrum of igneous rocks that occur on Earth, I asked a basic

question: what patterns, even with no knowledge of differentiation products? It would appear that the latter is the case and that there are at least three principal magma series, and perhaps several other minor ones.

Summary

Petrologists use the chemical composition of rocks and minerals to characterize and categorize individual types, to identify series of related rocks, and to model processes that relate the rocks in a consanguineous series. Analyses were initially performed by using tedious gravimetric methods but are now quickly and routinely accomplished by using instrumental spectroscopic techniques based on emission or absorption of energy when directed at prepared samples. Analyses are typically reported in wt. % oxide for the major

used, particularly ones in which various components are plotted versus some chemical parameter that is considered a measure of chemical fractionation (e.g., SiO_2 , Zr, Mg#). Ternary diagrams, such as the AFM diagram, are also common. By using bivariate diagrams, such as Harker or Pearce element ratio diagrams, we can evaluate specific differentiation mechanisms (e.g., fractional crystallization of certain minerals) deemed responsible for the chemical evolution within a suite of supposedly consanguineous rocks. Quanti-

Figure 3 (Winter Cross-Section of Subduction Magmatism): Dehydration of the subducting oceanic plate at 80–120 km depth releases aqueous fluids containing mobile LILE (Cs, Rb, Ba, Sr, K) and boron into the overlying mantle wedge (peridotite). The addition of water dramatically depresses the mantle peridotite solidus (flux melting), generating primary hydrous calc-alkaline picritic and basaltic melts that ascend into the island arc or continental crust.

*"Because the **calc-alkaline series is restricted mostly to convergent plate boundaries** in complex settings, we shall set it aside [when modeling oceanic basalt generation at mid-ocean ridges and intraplate islands]."*

— John D. Winter, *Principles of Igneous and Metamorphic Petrology*, Chapter 10, p. 191.

4. Cross-Correlation: Magma Series vs. Plate Tectonic Settings

The following matrix compares the four magma series across all major global tectonic regimes:

Magma Series	Subduction Zones (Island & Continental Arcs)	Divergent Margins (MORB / Mid-Ocean Ridges)	Oceanic Intraplate Hotspots (OIB / Hawaii)	Continental Intraplate Rifts & Flood Basalts
Calc-Alkaline ★ Found ONLY in Subduction	DOMINANT & DIAGNOSTIC (Andes, Cascades, Japan, Aleutians).	ABSENT (No water/fluxing at ridges).	ABSENT (Anhydrous plume melting).	ABSENT (Unless contaminated by ancient subduction-modified crust).
Tholeiitic Option (b)	Present in immature island arc tholeiites (IAT).	DOMINANT (Nearly 100% of normal MORB is tholeiitic).	DOMINANT (Huge shield-building lavas of Hawaii, Iceland, Galapagos).	DOMINANT (Continental Flood Basalts: Deccan Traps, Karoo, Columbia River).
Alkaline Option (c)	Rare / Minor (Late back-arc or post-collisional shoshonites).	Very Rare (Only localized off-axis seamounts or early embryonic rifts).	COMMON & CHARACTERISTIC (Post-shield rejuvenated stages: alkali basalt, phonolite).	COMMON & CHARACTERISTIC (Continental rifts: East African Rift, Oslo Graben).
Komatiitic Option (d)	Absent in modern subduction zones.	Absent (Requires mantle temperatures $>1600^{\circ}\text{C}$).	Absent in modern plate tectonics (Archean mantle plume origin, $>2.5\text{ Ga}$).	Archean greenstone belts (e.g., Barberton, Abitibi, Dharwar).



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 19

Which one of the following magma series is found only in subduction zones?

- (a) Calc-alkaline
- (b) Tholeiitic
- (c) Alkaline
- (d) Komatiite

Correct Option: (a) Calc-alkaline



Verified via John D. Winter, *Principles of Igneous & Metamorphic Petrology* (Ch. 8, p. 161; Ch. 10, p. 191; Ch. 16 & 17)

CORRECT ANSWER Option (a): Calc-alkaline

Detailed Validation: The **calc-alkaline magma series** is universally recognized in igneous petrology as being uniquely and exclusively generated at **subduction-related convergent plate margins** (island arcs and active continental margins, such as the Japanese Arc, Aleutians, and the Andes). The genesis of calc-alkaline magmas fundamentally requires high volatile (H_2O) content and elevated oxygen fugacity (f_{O_2}), which are provided solely by dehydration fluids expelled from the subducting lithospheric slab into the overlying mantle wedge. Under these conditions, early crystallization of Fe-Ti oxides (titanomagnetite) prevents iron enrichment and drives liquid differentiation toward silica-rich andesites, dacites, and rhyolites. Calc-alkaline magmatism does not occur at mid-ocean ridges, intraplate oceanic hotspots, or continental flood basalt provinces.

INCORRECT Option (b): Tholeiitic

Detailed Refutation: The *tholeiitic series* is the most widespread and ubiquitous magma type on Earth. Far from being restricted to subduction zones, tholeiitic basalt constitutes virtually 100% of Mid-Ocean Ridge Basalts (MORB—the largest magmatic province on the planet at divergent margins), dominates oceanic island shield volcanoes (e.g., Mauna Loa and Kilauea in Hawaii, Iceland), and forms massive Continental Flood Basalt provinces (e.g., Deccan Traps, Columbia River Basalts). Because tholeiites occur in nearly every tectonic environment, they cannot be diagnostic of subduction zones.

INCORRECT Option (c): Alkaline

Detailed Refutation: The *alkaline magma series* is characteristically associated with **intraplate tectonic settings**, specifically continental rifts (e.g., the East African Rift System, Rio Grande Rift) and late-stage oceanic island hotspots (e.g., the post-shield and rejuvenated stages of Hawaii, Tahiti, Tristan da Cunha). Alkaline magmas are generated by low degrees of partial melting of deep, garnet-bearing asthenospheric mantle in intraplate regimes, not in subduction zones.

INCORRECT Option (d): Komatiite

Detailed Refutation: *Komatiites* are ultra-magnesian ($>18\text{ wt}\% \text{ MgO}$), high-temperature ($>1600^\circ\text{C}$) volcanic rocks displaying spinifex textures. They were generated almost exclusively during the **Archean Eon** ($>2.5\text{ Ga}$) by exceptionally high-degree ($>40\%$) partial melting of deep mantle plumes. Komatiites are not a feature of modern plate tectonics or subduction zones.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering AFM Projections, Miyashiro Plots, Slab Fluids & Arc Petrochemistry

High-Yield Exam Practice Questions

Practice Q1 (Differentiation Trends): AFM Path Divergence

On an AFM ternary diagram, why does the Skaergaard intrusion liquid trend curve sharply upward toward the F apex, whereas the Crater Lake (Cascades) liquid trend moves straight toward the A apex?

Solution & Rationale:

Skaergaard is an anhydrous, reducing **tholeiitic** system where plagioclase + olivine fractional crystallization leaves iron to concentrate in the residual melt (Fenner trend of iron enrichment). Crater Lake is a hydrous, oxidized **calc-alkaline** arc magma where elevated f_{O_2} triggers early crystallization of titanomagnetite (Fe_3O_4), immediately removing iron and steering the remaining melt straight toward silica and alkalis ($\text{Na}_2\text{O} + \text{K}_2\text{O}$).

Practice Q2 (Subduction Geochemistry): The Tonalite-Trondhjemite-Granodiorite (TTG) Suite

In Archean subduction zones, what calc-alkaline plutonic suite formed the primary continental crust, and what mineral residue was required in the slab source?

Answer: The **TTG suite** (Tonalite–Trondhjemite–Granodiorite). It was formed by partial melting of hot subducting oceanic crust in the garnet amphibolite / eclogite facies, leaving a **garnet + rutile** residue that caused heavy REE depletion and high Sr/Y ratios.

Practice Q3 (Miyashiro Discriminant): Slope Significance

In Miyashiro's (1974) plot of FeO^*/MgO versus SiO_2 , what mathematical criterion separates tholeiitic suites from calc-alkaline suites?

Answer: The slope of the differentiation trend ($\Delta(\text{FeO}^*/\text{MgO}) / \Delta\text{SiO}_2$). Tholeiitic suites show a steep slope (> 0.1), indicating rapid increase in iron relative to magnesium with minimal silica change. Calc-alkaline suites have a shallow slope (< 0.1), indicating that silica increases rapidly as magnesium decreases, without an iron spike.

Practice Q4 (Subduction Volatiles): Slab Dehydration Mechanism

Which metamorphic mineral reactions in the down-going oceanic slab are primarily responsible for releasing the water that fluxes the mantle wedge at 100 km depth?

Answer: The breakdown of hydrous phyllosilicates and amphiboles during the transition from blueschist to eclogite facies—specifically the dehydration of **serpentine**, **chlorite**, **lawsonite**, and **glaucophane** into anhydrous garnet + omphacite + quartz.

Practice Q5 (Trace Element Signature): Arc Spidergrams

What is the most diagnostic geochemical trace element fingerprint of calc-alkaline subduction zone magmas on a primitive-mantle-normalized spidergram?

Answer: Enrichment in Large-Ion Lithophile Elements (LILE: $\text{Cs, Rb, Ba, Sr, Pb}$) coupled with pronounced negative anomalies in High Field Strength Elements (Nb, Ta, Ti). This "subduction signature" reflects the mobility of LILE in slab-derived aqueous fluids and the retention of Nb-Ta-Ti in residual slab rutile.

Practice Q6 (Petrography): Dominant Rock Types

What volcanic rock type is volumetrically dominant in calc-alkaline continental volcanic arcs (e.g., the Andes), contrasting with tholeiitic mid-ocean ridges where basalt dominates?

Answer: **Andesite** ($\text{SiO}_2 \approx 57\text{--}63\text{ wt}\%$), accompanied by dacite and rhyolite. The worldwide abundance of andesite at convergent margins gave the "Andes" its name and defines the "Andesite Line" around the Pacific Rim.

Practice Q7 (Thermal Divide): Alkaline vs. Subalkaline Immutability

Why can a subalkaline magma (tholeiitic or calc-alkaline) never differentiate into an alkaline magma (e.g., phonolite) during low-pressure fractional crystallization?

Answer: Because of the presence of the **thermal divide** (the plane of silica saturation: Forsterite–Diopside–Albite) in the basalt tetrahedron. Magmas on the silica-saturated side of the divide evolve toward the silica-oversaturated minimum (quartz-normative rhyolite), while magmas on the undersaturated side evolve toward the feldspathoid minimum (nepheline-normative phonolite). Crystallization cannot cross a thermal divide.

Practice Q8 (Shoshonitic Series): Arc Evolution

In mature island arcs and post-collisional settings, an ultra-potassic magma series often appears behind the volcanic front. What is this series called?

Answer: The **Shoshonitic Series** (or high-K calc-alkaline to potassic alkaline series). It represents magmas derived from deep, metasomatized mantle lithosphere enriched in potassium and LILE during waning or terminated subduction.

Comprehensive Magma Series Summary Matrix

Magma Series	Parental Magma	Differentiation Trend	Volatiles & Redox State	Tectonic Setting
Calc-Alkaline (The Answer)	Hydrous High-Alumina Basalt	Silica enrichment without Fe enrichment (Andesite $\rightarrow$ Dacite $\rightarrow$ Rhyolite)	Hydrous ($>2-6\text{ wt}\% \text{H}_2\text{O}$); High f_{O_2} (NNO to HM buffer)	Exclusively Subduction Zones (Island Arcs & Active Continental Margins)
Tholeiitic	Olivine Tholeiite Basalt	Iron enrichment (Fenner trend) followed by silica enrichment	Anhydrous ($<0.5\text{ wt}\% \text{H}_2\text{O}$); Low f_{O_2} (QFM buffer)	Mid-Ocean Ridges (MORB), Ocean Islands (OIB), Continental Flood Basalts, Back-Arcs
Alkaline	Alkali Olivine Basalt / Basanite	Silica-undersaturated trend (Trachybasalt $\rightarrow$ Trachyte $\rightarrow$ Phonolite)	Variable, enriched in CO_2 and alkalis	Intraplate Hotspots (Hawaii, Tahiti), Continental Rifts (East African Rift)
Komatiitic	Peridotitic / Ultramafic Komatiite	Olivine fractionation from ultra-high-Mg liquids ($>18\text{ wt}\% \text{MgO}$)	Anhydrous, extremely high temperature ($>1600^\circ\text{C}$)	Archean Greenstone Belts (Archean Mantle Plumes, $>2.5\text{ Ga}$)